

OFDG: a laptop-sized numerical preview

Initial evidence for choosing the next larger experiments

This brief compares accuracy, shock resolution, and positivity on small paper-inspired tests. It supports a supervisor discussion; it does not establish a universal method ranking.

What the completed runs suggest

Smooth accuracy: on the finest P2 grids, KXRCF gating keeps errors close to DG; ungated OFDG has larger errors.

Scalar shocks: filtering suppresses the large sampled DG excursions. The two OFDG variants are often nearly coincident.

Next decision: use Shu-Osher wave preservation and curved geometry to distinguish the methods more carefully before scaling up.

Experiment	Configuration	Question
Smooth advection	P1-P3; N=32/64/128/256; T=1.1	Accuracy across order
Piecewise advection	P2; N=128/256; T=1.1	Oscillations and smearing
Burgers	P2; smooth N=32/64/128, T=0.6; shock N=128/256, T=2.2	Nonlinear shock formation
Lax	P1/192, P2/128, P3/96; T=1.3	Equal unknown count
Shu-Osher	P2; N=200/400; T=1.8	Shock-wave interaction
Straight / curved transport	P2; 8 ² /16 ² /32 ² cells; T=1	Geometry and accuracy
2D Riemann	P2; 32 ² /64 ² cells; T=0.25	Wave structure and positivity

Methods and notation

DG is discontinuous Galerkin without the oscillation filter. **OFDG** is its oscillation-free variant; **OEDG** is the oscillation-eliminating comparator. **KXRCF gating** applies OFDG only in cells flagged by a troubled-cell indicator. Pk denotes polynomial degree k; N is the 1D cell count. Squared counts specify 2D grids.

All runs use RK4. CFL is 0.15 for scalar problems and 0.30 for Euler. OFDG filters after each full step; OEDG filters at RK stages. Every Euler method, including DG, uses the same positivity limiter. OEDG appears on selected finer affine shock cases only.

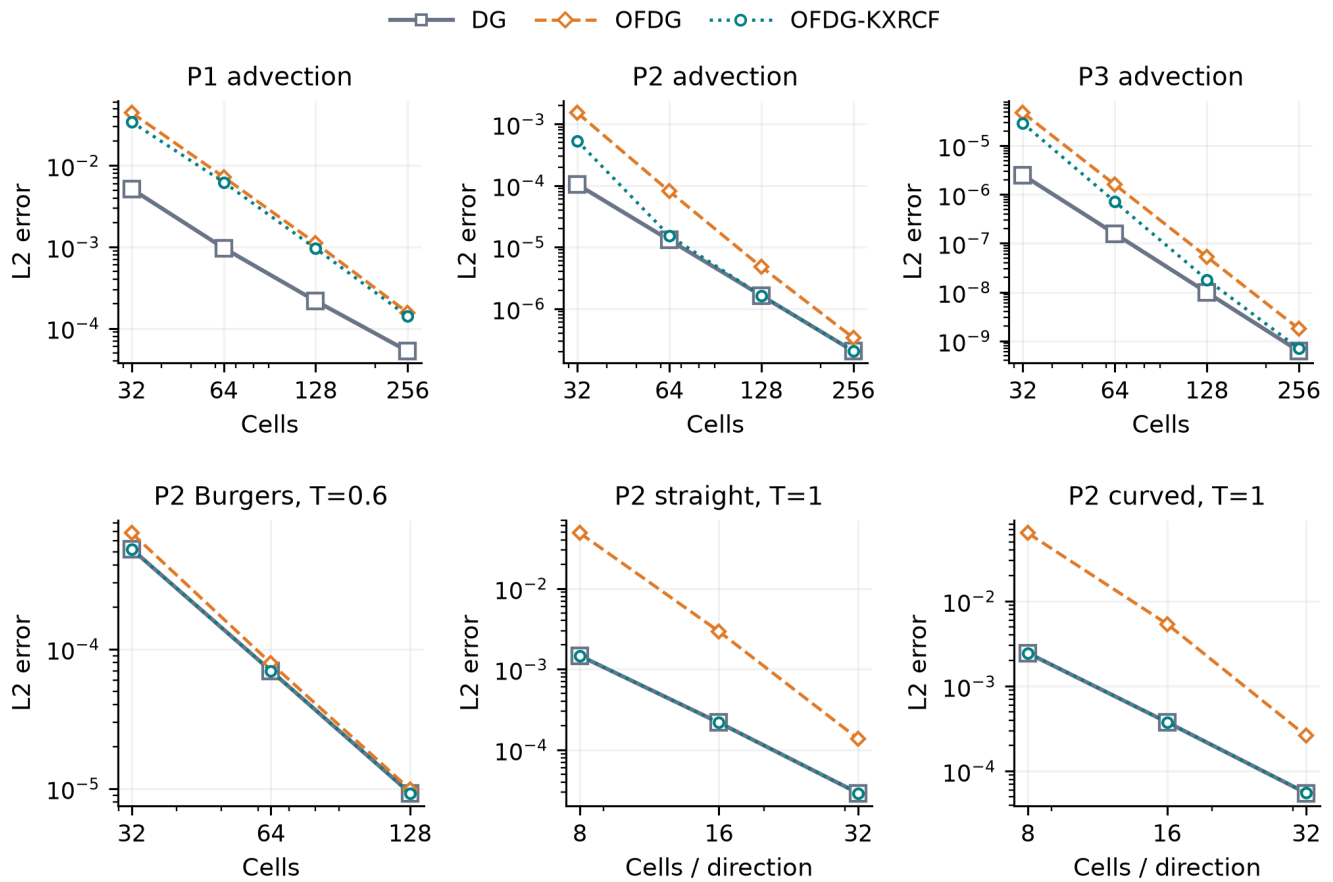
Compute used

103 solver runs and 4 reference solves completed in 7.35 minutes, with sampled peak process-tree memory 118 MiB. Runs were sequential and used at most two MPI ranks. No cases failed or were dropped.

Ceilings: 4 GB, five minutes per check, 60 minutes total; no new checks after 45 minutes. Memory peaks are sampled. Build and document preparation are excluded from solver time. Single-run timings do not establish performance differences.

Smooth accuracy and curved geometry

Refinement reduces every smooth error. Higher slopes in the filtered cases on these grids do not establish a higher asymptotic order.



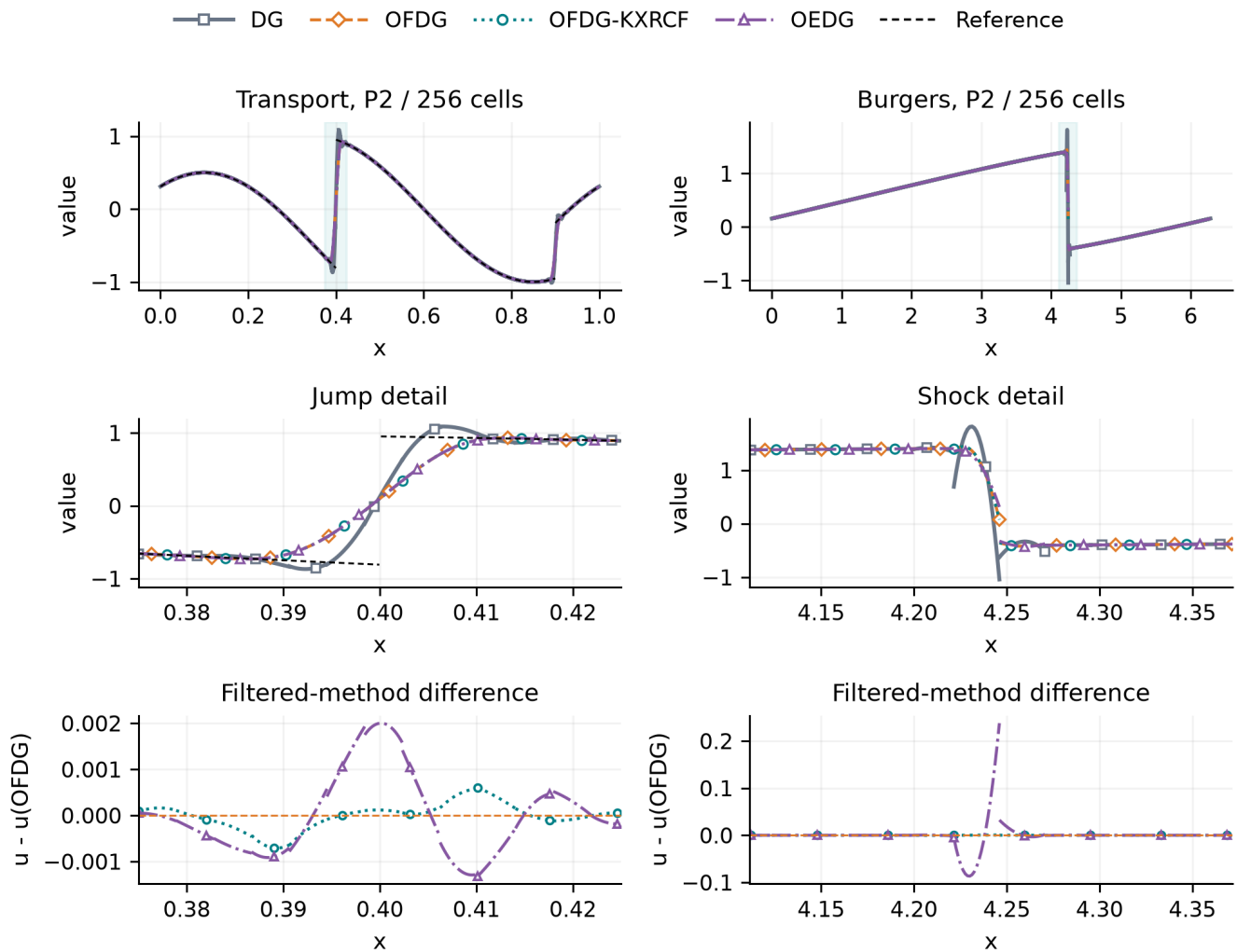
Method	P1 slope	P2 slope	P3 slope	Curved slope	P2 error / DG	Curved error / DG
DG	2.05	3.00	4.00	2.77	1.000	1.000
OFDG	2.83	3.83	4.90	4.37	1.648	4.715
OFDG-KXRCF	2.77	3.00	4.70	2.77	1.000	1.002

Slopes use the finest two grids. Nominal spatial orders are 2, 3, and 4 for P1, P2, and P3. Error ratios use $N=256$ (1D P2) and 32^2 (curved P2). Nested markers indicate nearly coincident curves: the table quantifies the hidden DG / gated-OFDG difference.

The straight and curved 2D panels share the unit-square transport problem: velocity $(0.7, 0.3)$, $\sin^2(\pi(x+y))$ data, and a static cubic deformation for the curved case. This is a 2021-paper-inspired adaptation.

Finest-P3 CFL halving changed L2 error by DG 0.0006%, OFDG 0.0029%, OFDG-KXRCF 0.0003%. No change exceeded 5%, so no extra series was needed. Maximum reported periodic scalar conservation drift: $4e-14$.

Scalar shocks: separate near-coincident curves



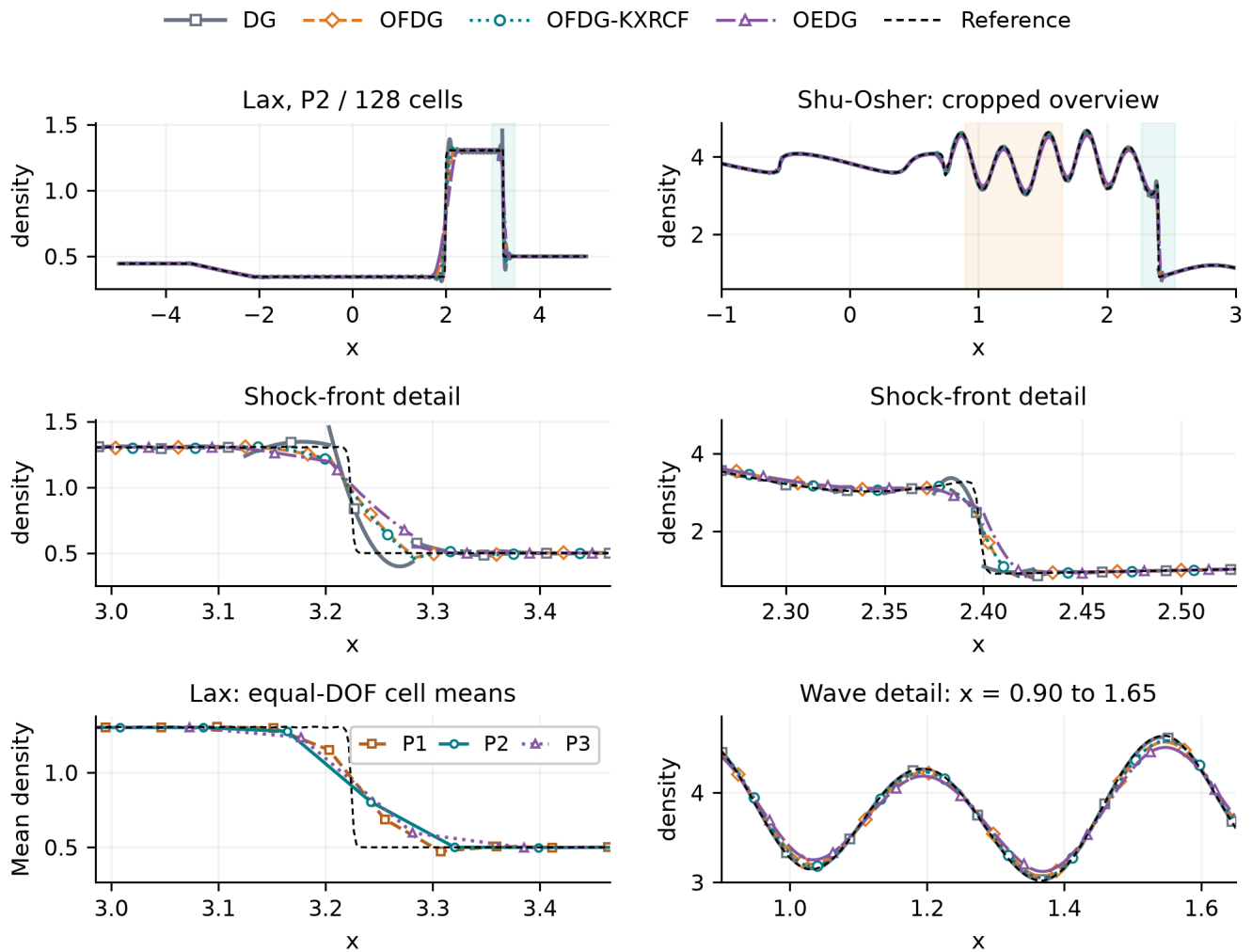
Overviews locate the shaded close-ups. The bottom row subtracts OFDG, so small differences between the filtered methods remain visible. The black reference is exact translated transport; Burgers has no reference curve.

Sampled excursion	Transport under	Transport over	Burgers under	Burgers over
DG	0.012	0.136	0.541	0.317
ODFG	$<10^{-6}$	$<10^{-6}$	$<10^{-6}$	$<10^{-6}$
ODFG-KXRCF	$<10^{-6}$	$<10^{-6}$	$<10^{-6}$	$<10^{-6}$
OEDG	$<10^{-6}$	$<10^{-6}$	$<10^{-6}$	$<10^{-6}$

Excursions are beyond initial-data bounds: transport $[-1, 0.951057]$, Burgers $[-0.5, 1.5]$. Values below 10^{-6} are grouped for display; full precision is retained in analysis.json. The plots use 21 samples per cell, with curves broken at DG interfaces; sampled extrema are not certified continuous extrema.

In the displayed windows, $\max |\text{gated OFDG} - \text{ODFG}|$ is 0.00071 for transport and 0.00049 for Burgers. Near coincidence is a result here, not evidence that the curves were omitted.

Euler: shock fronts and wave detail



Blue bands mark shock details; the orange band marks the Shu-Osher wave detail. Shu-Osher uses P2 / 400 cells; its overview is cropped to [-1,3] from the [-5,5] domain. The equal-DOF Lax panel uses gated OFDG with P1/192, P2/128, P3/96: 384 scalar unknowns each. Its markers are cell means, not polynomial samples.

Wave-window peak-to-trough density range: DG 1.590, OFDG 1.495, OFDG-KXRCF 1.529, OEDG 1.383, WENO 1.615. This local range is a first look at amplitude; it does not measure phase accuracy.

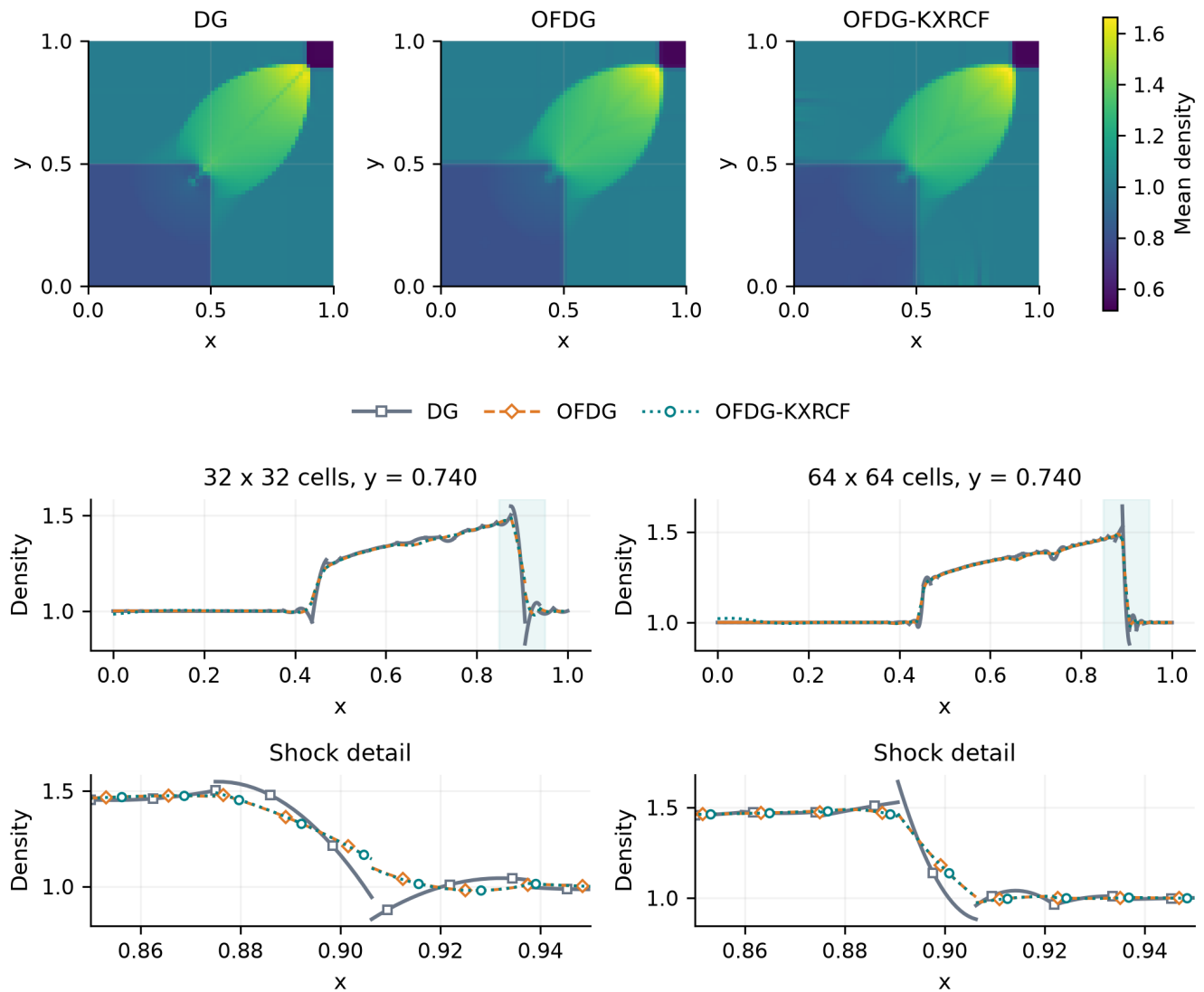
WENO reference pair	Mean absolute difference	Maximum difference
Lax 2048 vs 4096	0.00101	0.12
Shu-Osher 2048 vs 4096	0.00324	0.463

These are density differences after averaging the finer WENO reference onto the coarser grid. Large localized differences near steep fronts limit pointwise judgments even when mean differences are small. WENO5 / SSPRK3 is independent of DG, but is not exact.

Shu-Osher positivity cell-limit events: DG 6252, OFDG 3, OFDG-KXRCF 3, OEDG 2. These are cumulative interventions, not distinct cells. Fixed far-field boundaries permit changes in domain totals; those changes are not conservation-error estimates.

2D Riemann: compare the same physical cut

P2 Euler, $T=0.25$; four-quadrant data with outflow boundaries. Maps show 64^2 physical cell means on one shared scale. Its range spans all displayed means, so no values are clipped.



Both grids are evaluated at exactly $y=0.740$ by reconstructing each P2 polynomial from its tensor-node samples. Curves remain broken at cell interfaces. The two resolutions are separated to keep method comparisons readable.

The two OFDG traces nearly coincide. Max $|\text{gated OFDG} - \text{OFDG}|$ along this cut: 0.0173 on 32^2 and 0.0226 on 64^2 . The close-ups use polynomial traces; the maps above use cell means.

Method / cells	Min. density	Min. pressure	Limit events	Rejected steps
DG / 32^2	0.38	0.207	122	0
OFDG / 32^2	0.513	0.379	0	0
OFDG-KXRCF / 32^2	0.513	0.379	0	0
DG / 64^2	0.431	0.236	482	0
OFDG / 64^2	0.515	0.384	0	0
OFDG-KXRCF / 64^2	0.515	0.384	0	0

Minima are the solver-reported final values. Limit events count cumulative cell interventions. All three methods stayed admissible with the common positivity treatment; these coarse grids do not establish a resolution-independent ranking.

Decisions for the next compute allocation

Priority	Evidence from this preview	Next experiment / decision
1	Shu-Osher filters look close; amplitude alone cannot establish phase or wave accuracy.	Refine the wave region; measure amplitude and phase against a checked reference. Repeat timings only for this selected case.
2	Gating nearly matches DG smooth errors; curved errors are larger on the same grid counts.	Refine matched straight / curved problems, then a small curved Euler case. Separate geometry error from filtering.
3	2D OFDG variants differ little on the sampled cut, while DG uses more positivity interventions.	Refine the Riemann problem with boundary conditions and limiter settings matched across future comparisons.
Later	No result here requires a giant test to interpret.	Defer blast waves, large 3D studies, and scaling studies until the scientific question and hardware are chosen.

What remains uncertain

This is a small configured comparison, not a full paper reproduction. Fluxes, initialization, filtering cadence, and positivity treatment affect the outcome. Only smooth P3 advection has a time-step sensitivity check. Shock widths and the wave-detail range need broader refinement. Native NURBS is outside the preview; curved results use supported polynomial geometry.

Open the evidence

Run: **preview**, 103 successful solver runs and four references.
Dataset: **measurements/study/preview/**
Instructions: **docs/supervisor-preview.md**
Configuration: **experiments/preview_manifest.json**

Paths are relative to `codex_dev`. The dataset retains exact commands, logs, physical samples, reference CSVs, convergence tables, and source/executable fingerprints. Existing thesis assets are unchanged. Project revision `f50a6dff8f98`; MFEM `a1ce49fb5742`.

Research basis

[1] J. Lu, Y. Liu, C.-W. Shu (2021). An Oscillation-Free Discontinuous Galerkin Method for Scalar Hyperbolic Conservation Laws. Scalar advection and Burgers examples. DOI: [10.1137/20M1354192](https://doi.org/10.1137/20M1354192)

[2] Y. Liu, J. Lu, C.-W. Shu (2022). An Essentially Oscillation-Free Discontinuous Galerkin Method for Hyperbolic Systems. Lax, Shu-Osher, and equal-DOF comparisons. DOI: [10.1137/21M140835X](https://doi.org/10.1137/21M140835X)

[3] M. Peng, Z. Sun, K. Wu (online 2024; issue 2025). OEDG: Oscillation-Eliminating Discontinuous Galerkin Method for Hyperbolic Conservation Laws. First four-quadrant Riemann setup. DOI: [10.1090/mcom/3998](https://doi.org/10.1090/mcom/3998)

Curved derivatives: does order matter?

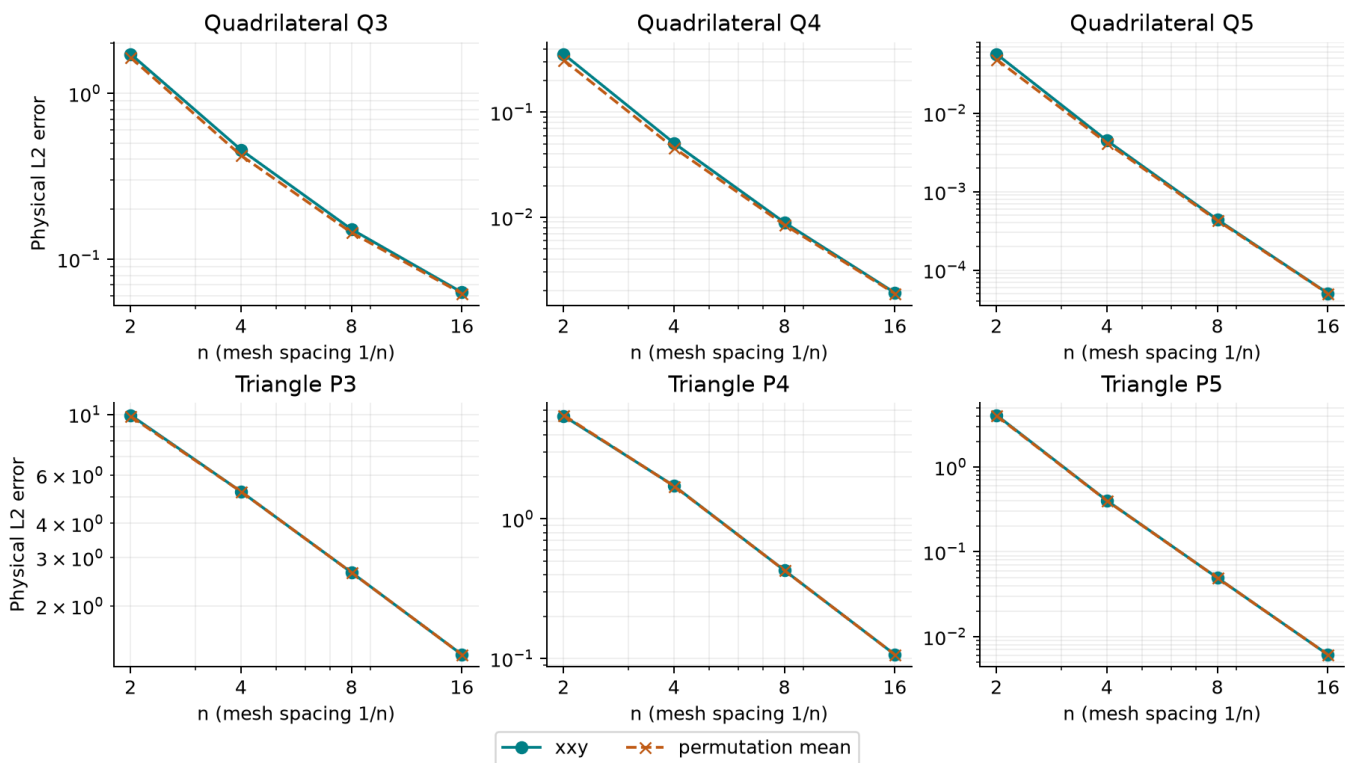
Isolated manufactured-derivative experiment; no PDE evolution and no change to the production OFDG algorithm. Exact mixed partial derivatives commute. This check measures the effect of inserting a physical L2 projection after each differentiation.

Construction and comparisons

Domain and geometry: start from (s,t) in $[0,1]^2$; set $b = a s(1-s)t(1-t)$, $x = s+b$ and $y = t+0.7b$. Use exact degree-4 polynomial geometry with $a = 0, 0.4, 1.2$ (affine, mild and stronger curvature). The domain boundary stays fixed; $\det J$ is at least $1-0.425a > 0$.

Spaces and data: triangle P3-P5 and quadrilateral Q3-Q5; $n = 2, 4, 8$, plus $n = 16$ at $a = 1.2$. Triangles have $2n^2$ elements, quadrilaterals n^2 ; these are not equal-DOF comparisons. Physically L2-project $\sin(2x+3y)$, $\exp(x+y/2)$, and x^2y+xy^2 into each space before differentiating.

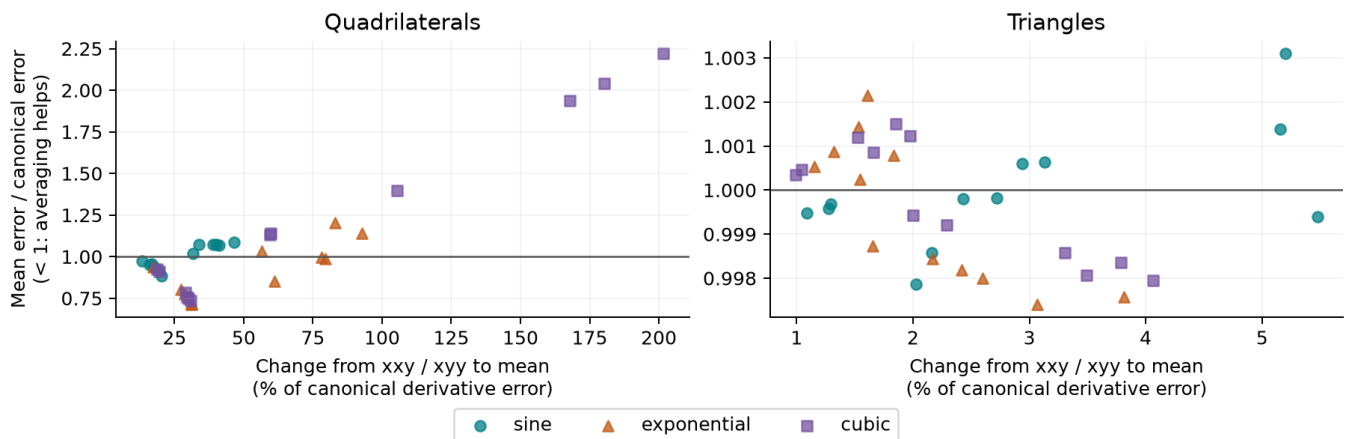
Sequences: words denote chronological operations: xy means x , then x , then y . Test xy/yx ; $xy/xyx/yxx$; and $xy/yxy/yxx$. The canonical sequence matches the production derivative graph. Average all distinct permutations as functions, then measure physical L2 errors against the analytic derivative. The average is a candidate approximation, not a reference solution.



Refinement: sine field, $a = 1.2$, xyx derivative. The exact derivative is $-12 \cos(2x+3y)$. Solid circles show the production order; dashed crosses show the permutation mean. Near-overlap is real; the next page plots changes directly. Both approaches improve under refinement in all six configurations.

Errors include differentiation of the initial solution projection error as well as the intermediate derivative projections. A separate direct projection of the analytic derivative supplies an approximation floor in the raw data. This is not a comparison with the exact higher derivative of the represented DG function.

Averaging: a change, not a guaranteed gain



Each point is one curved $n=8$ case: P/Q3-P/Q5, both nonzero curvature amplitudes, and xy or xyy . Horizontal position measures the L2 change to the derivative, divided by its canonical error. Vertical position measures the resulting error ratio. Points above 1 became less accurate after averaging. Axes use different scales: the triangle accuracy changes are much smaller.

Sine, $a=1.2$, $n=8$	Canonical error	Mean error	Error change
Q3 / xyy	0.1515	0.1446	-4.56%
Q4 / xyy	0.008933	0.008491	-4.95%
Q5 / xyy	0.0004397	0.0004283	-2.59%
P3 / xyy	2.649	2.649	-0.02%
P4 / xyy	0.4284	0.4278	-0.14%
P5 / xyy	0.04932	0.04922	-0.21%

Across 72 curved $n=8$ third-derivative cases, averaging lowers the error in 41. Mean/canonical error ratios range from 0.712 to 2.220 (median 0.999). The derivative change itself ranges from 1.0% to 201.6% of canonical error.

The largest worsening is mild-curvature Q3, cubic field, xyy : error 0.00109 to 0.00242 (exact derivative = 2). The relative errors are only 0.055% and 0.121%; the factor alone exaggerates its absolute significance.

Controls and interpretation

Affine canonical-to-mean differences are at most $5.6e-09$ in L2; the affine cubic derivative error is at most $3.3e-09$. Independent first-derivative weak residuals are below $3.4e-15$; constant-derivative coefficients below $8.5e-12$. Raising assembly quadrature by eight on four strong-curvature cases changes reported error norms by at most $2.4e-05$ relative.

Decision: retain the canonical implementation for now. Averaging removes the arbitrary ordering convention, but does not reliably reduce derivative error. These volume norms do not measure OFDG face-jump rates, damping, or final PDE accuracy. A targeted face-sensor and smooth-advection comparison is the next step before changing the method.

Resources and evidence: 64 sequential checks, one rank/thread; 4.63 minutes total, sampled peak 19.9 MiB. No new large PDE runs. Raw data, commands, hashes, quadrature controls and derived tables: [measurements/derivative-permutations/](#). Reproduction: [docs/derivative-permutations.md](#). This addendum is separate from the thesis.

How accurate are the derivatives?

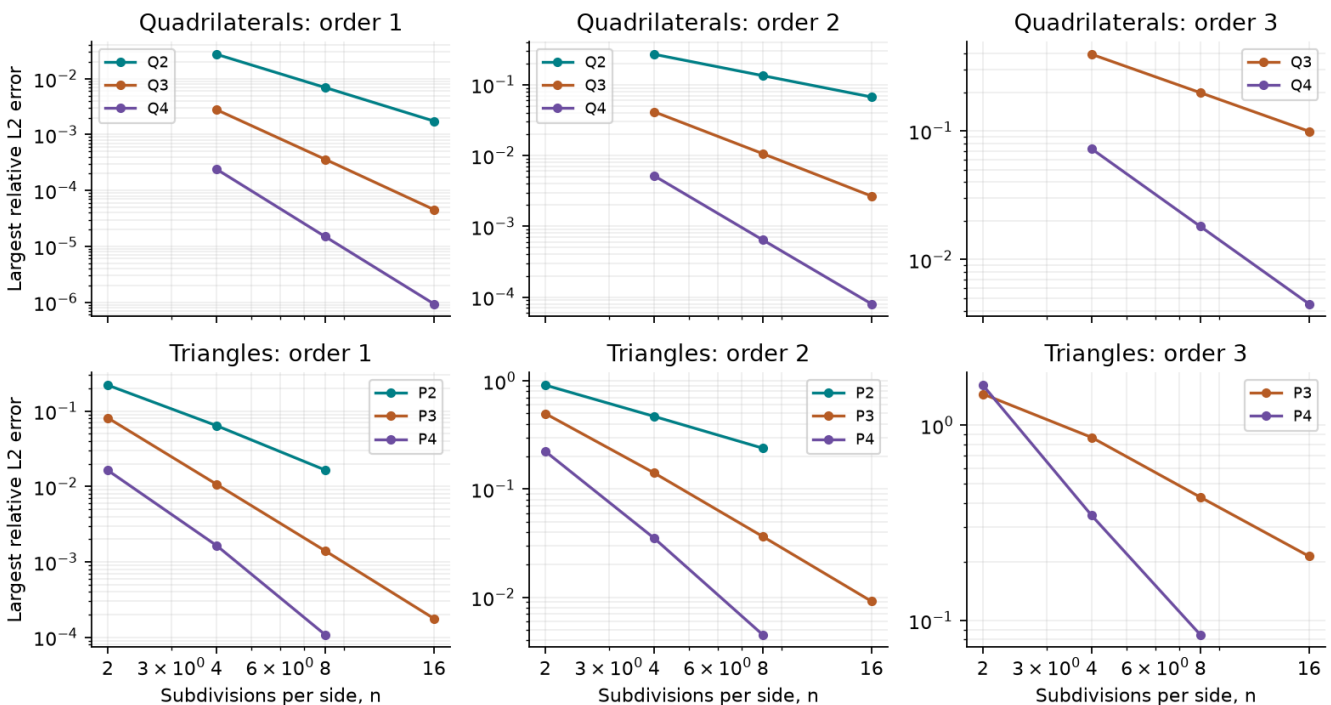
An isolated operator study compares the production projected-derivative matrices with analytic physical derivatives. No PDE is advanced and no production filtering code is changed. Higher derivative errors can be substantial even when first derivatives are accurate.

Two questions, tested separately

A. Complete approximation: physically project $\sin(2x+3y)$, $\sin(8x+6y)$, and $\exp(x+y/2)$, then differentiate recursively. Errors include the input projection and all intermediate derivative projections. Use the coupled curved-face map from page 7, with $a=0$, 0.4, 1.2.

B. Derivative projection alone: use an exactly representable mapped quadratic and an analytically invertible non-affine map (next page). This removes input approximation error and permits comparison with exact derivatives of the represented DG function itself.

Coverage: P/Q1-P/Q4 for A; P/Q2-P/Q4 for B. All canonical coordinate derivatives through $\min(p,3)$: $x, y, xx, xy, yy, xxx, xxy, xyy, yyy$. Gauss-Lobatto, Gauss-Legendre and positive Bernstein bases. Refinement uses $n=4,8,16$ for quads and $n=2,4,8$ for triangles, with selected $n=16$ triangle checks. Basis comparisons use $n=8$; the raw manifest identifies every case. Geometry degree 4 represents both maps exactly. Spaces at equal n are not equal-DOF comparisons.



Complete-approximation errors: $\sin(2x+3y)$, strong coupled curvature $a=1.2$, Gauss-Lobatto basis. Each curve shows the largest relative volume L2 error among derivatives of the indicated order. Different panel scales expose the refinement trend; high derivative orders remain much less accurate. These are canonical sequences, not permutation averages.

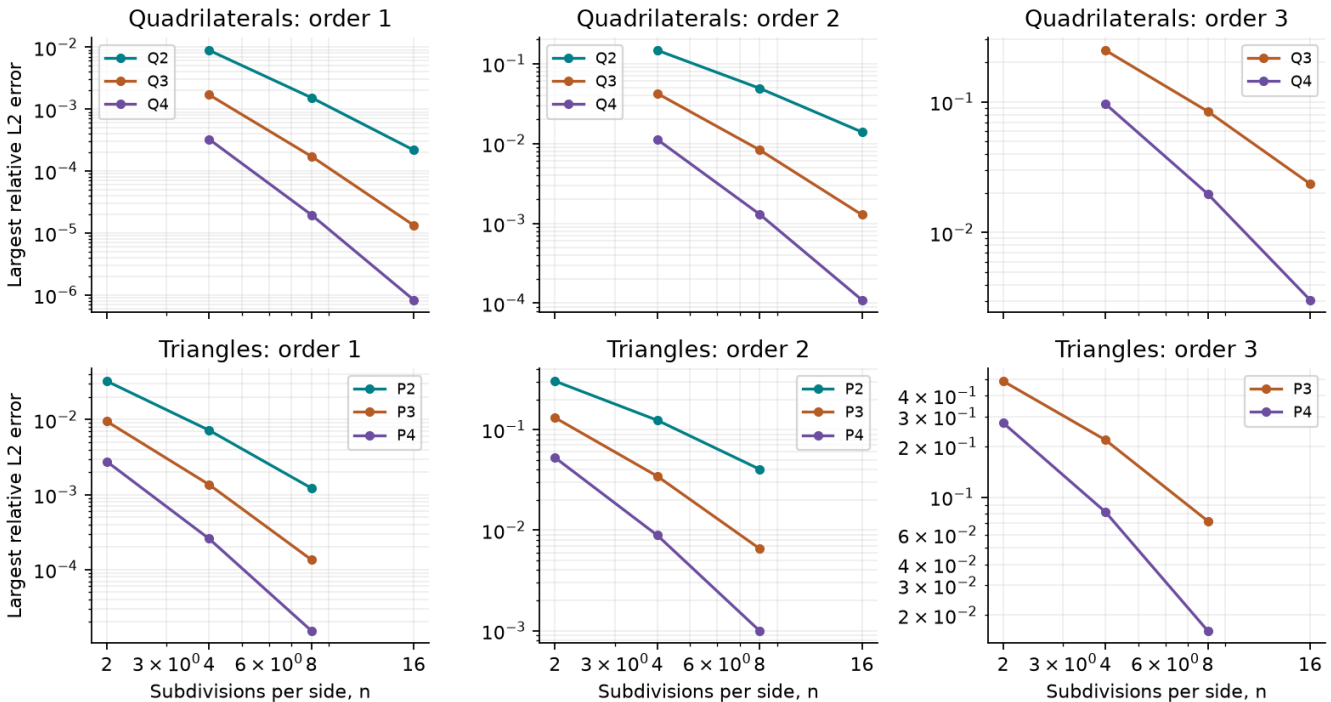
What “orders of magnitude off” means here

Absolute error $E = \|D^{\alpha} u_h - \text{partial}^{\alpha} u\|_{L_2}$; relative error $R = E / \|\text{partial}^{\alpha} u\|_{L_2}$. Thus $R=10^{-2}$ means a 1% norm error, two orders below the derivative norm. It is not a pointwise guarantee of two correct digits. Zero analytic derivatives have undefined relative error and are reported with absolute errors. No regularizing denominator hides those cases.

Separate projection error from input error

For the isolation test, take $x=s+a s(1-s)$, $y=t+0.7a t(1-t)$, and $u(x,y)=s(x)^2+s(x)t(y)+t(y)^2$. Test $a=0.3$ and 0.7 . On every element its pullback is quadratic, so P/Q2 and above represent it exactly. Its physical derivatives are generally nonpolynomial. The inverse map and its first three derivatives are analytic; see the reproducibility note.

The separable map stretches coordinate lines nonlinearly. Quad edges stay straight, while triangle diagonals can curve; it isolates non-affine parameterization rather than reproducing the curved quad faces of study A. Both maps are nonsingular throughout the domain.



Projection-only errors: strong separable map $a=0.7$. Initial physical projection error is at most $1.3e-14$ in L2 across the tested cases. All plotted errors can therefore be attributed to derivative representation, recursion and numerical roundoff, rather than an unresolved input field.

n=8, strong map	First derivative	Second derivative	Third derivative
Q2	3.08e-03 0.153% (x)	9.22e-01 4.91% (xx)	not used
Q3	3.47e-04 0.0173% (x)	1.56e-01 0.831% (xx)	5.25e+01 8.46% (xxx)
Q4	3.95e-05 0.00196% (x)	2.44e-02 0.13% (xx)	1.23e+01 1.98% (xxx)
P2	2.46e-03 0.122% (x)	7.58e-01 4.04% (xx)	not used
P3	2.73e-04 0.0136% (x)	1.23e-01 0.655% (xx)	4.48e+01 7.22% (xxx)
P4	3.08e-05 0.00153% (x)	1.87e-02 0.0997% (xx)	1.00e+01 1.61% (xxx)

Each cell gives absolute L2 error, then relative error; the displayed derivative has the largest relative error at that order. The worst absolute-error direction may differ. A direct physical projection of each exact derivative supplies the best-approximation floor. Higher recursive errors can exceed this floor because earlier projection errors are differentiated again. The raw results include both the floor and the excess over it.

Implications for OFDG and checks

How large are complete-approximation errors?

n=8; a=1.2	Order 1	Order 2	Order 3	Worst face error
Q2 / sine	1.48e-02 0.695%	3.78e-01 13.4%	not used	18.6%
Q3 / sine	7.67e-04 0.0361%	6.73e-02 1.06%	1.12e+00 19.8%	27.6%
Q4 / sine	3.17e-05 0.00149%	4.07e-03 0.0641%	1.03e-01 1.81%	3.2%
P2 / sine	2.34e-02 1.65%	6.73e-01 23.9%	not used	31%
P3 / sine	1.98e-03 0.14%	1.03e-01 3.64%	2.43e+00 42.8%	56.1%
P4 / sine	1.53e-04 0.0108%	1.27e-02 0.45%	4.80e-01 8.46%	13.8%

Volume cells show absolute L2 error / largest relative error over the canonical derivatives at each order. The last column is the largest relative one-sided face L2 error over all tested derivative orders, normalized by the analytic trace norm. It uses every element boundary, counting interior faces twice. It is a trace diagnostic, not the OFDG jump or damping rate.

Across these n=8 P/Q2-P/Q4 strong-curvature cases, the order-wise worst relative errors range from 2.6e-07 to 1.9 for the sine/exponential fields, and 0.00049 to 0.57 for the shorter wave. These are envelopes of a configured test set, not a statistical statement about typical meshes. Sampled maximum errors and nominal-h-scaled errors are also saved.

Basis, integration and analytic controls

Changing among Gauss-Lobatto, Gauss-Legendre and Bernstein bases changes the reported L2 error norms by at most 5e-08 absolute (8.4e-10 on nonconstant fields after scaling by the larger analytic derivative/input norm). This compares error norms, not pointwise equality of the derivative functions. Independent first-derivative weak residuals are at most 3.9e-15. Constant-field sampled derivative magnitudes reach 1.8e-06: repeated differentiation amplifies tiny initialization/assembly roundoff. Relative errors for these zero derivatives are undefined.

Raising assembly quadrature by eight changes error norms by at most 1.8e-10 on that scale; independently raising initialization/evaluation quadrature by eight changes them by 3.2e-10. Evaluation rules otherwise use production order +16. Analytic inverse-map derivatives are checked independently by high-precision finite differences. Sampled maxima are not certified continuous suprema.

Assessment: useful, with a resolution-dependent boundary

Retain this construction for OFDG, but do not describe its higher derivatives as uniformly accurate. Refinement and increased order improve the tested errors; changing coefficient basis does not cure them. First derivatives can be several orders below the true derivative norm while the highest derivative used by a low-order space remains at percent-to-tens-of-percent error. Strong distortion and shorter waves require particular care.

OFDG uses derivatives as roughness indicators, not as a replacement for the DG flux. That makes this a sensible construction to continue evaluating, but volume accuracy alone cannot establish acceptable damping or PDE accuracy. Face errors can be larger. Before claiming the extension is sufficiently accurate, compare face-jump rates and smooth-advection convergence against an independent derivative construction; shock robustness requires a separate check.

Resources: 135 completed checks, 0 failures; one rank/thread on an 8-logical-processor laptop. 14.69 minutes charged compute (including a conservative interruption allowance), peak sampled RSS 21.3 MiB. Raw data, exact commands, controls, source hashes and tables: [measurements/derivative-accuracy/](#). Instructions: [docs/derivative-accuracy.md](#). No new shock/PDE runs; no claim of stability or a universal error bound.

Differentiate the original mapped DG function

Question: can we evaluate physical derivatives of the existing DG function directly, avoiding every intermediate projection? This is an isolated operator experiment. The production OFDG implementation and all earlier raw datasets remain unchanged.

What changes, and what stays the same?

Current recursive projection	Direct mapped differentiation
After each physical derivative, project back into the original mapped DG space. Repeat on the projected coefficients.	Differentiate the original mapped function through the curved coordinate map. Evaluate the requested derivative directly at quadrature points.
Intermediate projections can change higher derivatives and make different derivative orderings disagree.	No derivative projection or new solution space. Mixed partials commute mathematically within each smooth, invertibly mapped element.

Both methods receive exactly the same physical L2-initialized DG coefficients. Direct differentiation does **not** reconstruct a polynomial in physical coordinates and does not use analytic solution derivatives to calculate the numerical answer. Analytic derivatives are used only as error references.

Mathematical construction

Write the reference solution as $u\text{-hat}(\xi) = \sum_j c_j \phi\text{-hat}_j(\xi)$, and the physical function as $u_h(x) = u\text{-hat}(F^{-1}(x))$. With $B = J^{-1}$, physical differentiation in direction i is $L_i = \sum_a B_{ai} \partial/\partial \xi_a$. Repeated application must differentiate B as well as the reference polynomial.

Equivalently, at a point x_q form the third-order Taylor expansion of $F^{-1}(x_q + h z)$. Compose the original reference polynomial with that expansion. Its coefficient of $z_x^a z_y^b$, multiplied by $a! b! / h^{a+b}$, gives the physical derivative with a x -differentiations and b y -differentiations. Truncation to degree three retains all derivatives through degree three; this is not a finite-difference approximation.

Implementation and support boundary

The prototype uses truncated bivariate Taylor arithmetic and four formal inverse-map corrections with the base-point Jacobian. Reference polynomials are recovered in a local monomial basis by interpolation, an algebraic change of basis checked against MFEM values and first derivatives. The known polynomial geometry formula is used directly and checked against MFEM point evaluations. No global inverse-map formula is required.

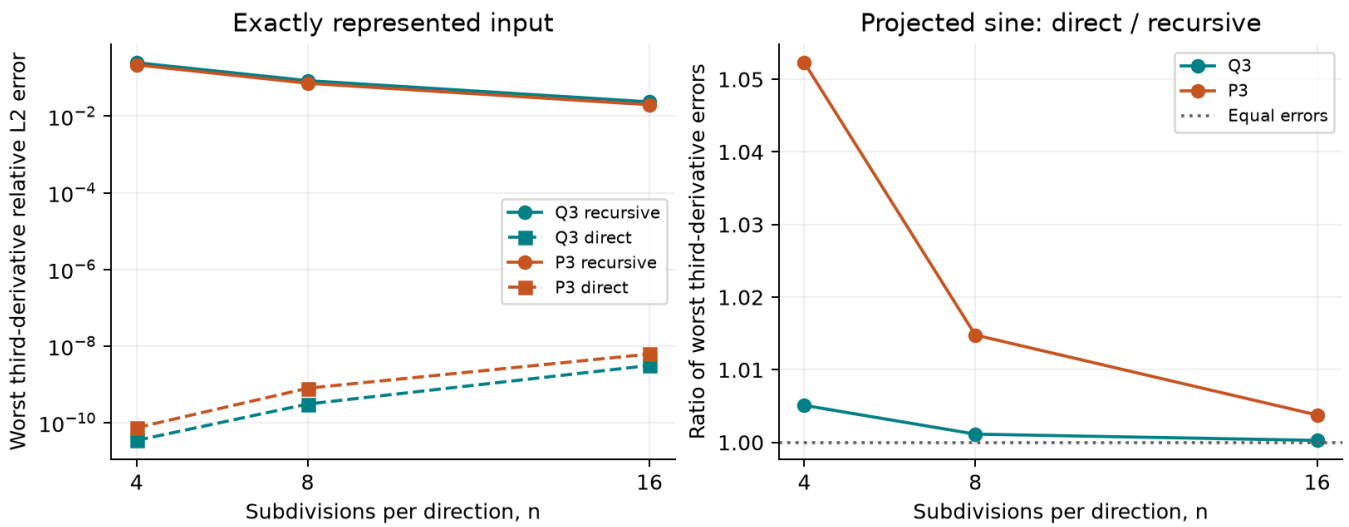
Coverage: polynomial 2D triangles and quadrilaterals, P/Q2-P/Q4, derivatives through $\min(k,3)$, and Gauss-Lobatto, Gauss-Legendre and Bernstein coefficient bases. These tests do not supply a general MFEM higher-geometry-derivative interface, NURBS/3D support, a cached production implementation, or a PDE stability result. Different derivative permutations are implicit in one Taylor coefficient; separate order-by-order implementations were not compared.

What improves, and what does not

Matched-input third derivatives: $n=8$, solution order $k=3$

Input / space	Recursive relative error	Direct relative error	Recursive / direct absolute error
Exact input / Q3	8.464%	1.833e-10%	52.5 / 1.14e-09
Exact input / P3	7.224%	1.376e-09%	44.8 / 8.54e-09
Projected sine / Q3	19.82%	19.84%	1.12 / 1.13
Projected sine / P3	42.83%	43.47%	2.43 / 2.47

Every table entry refers to the same xxx derivative. Relative error is $\| \text{numerical derivative} - \text{analytic derivative} \|_{L_2(K \text{ summed})} / \| \text{analytic derivative} \|_{L_2(K \text{ summed})}$. Absolute errors are physical volume L2 norms. Exact-input third derivatives can be large; percentages and absolute errors therefore answer different questions.



Plots use the largest relative error over xxx, xxy, xyy, yyy; the maximizing direction can differ between methods. Left: direct errors are limited by amplified floating-point/input roundoff, so their increase under refinement is not a truncation-error convergence rate. Right: ratios above one mean direct differentiation has a larger error. Ratios expose differences that would be hidden by overlapping error curves.

Why these outcomes coexist

Direct differentiation removes the recursive operator error relative to the represented u_h . It does not remove the differentiated input error, $\text{partial}^{\text{alpha}}(u_h - u)$. For the exactly represented input, that term vanishes in exact arithmetic and the improvement is dramatic. For the projected sine, differentiation of the existing DG approximation dominates; removing intermediate projections gives little benefit and can slightly increase error. Projection can also suppress components of the input error. No general ranking against the analytic solution follows.

Diagnostic fields and geometry

Original coordinates (s,t) cover the unit square, with n^2 quads or $2n^2$ triangles. Sine: $u = \sin(2x+3y)$, $x=s+b$, $y=t+0.7b$, $b=1.2s(1-s)t(1-t)$. Exact input: $u=s(x)^2+s(x)t(y)+t(y)^2$, $x=s+0.7s(1-s)$, $y=t+0.49t(1-t)$. The latter is a non-affine map with straight quad edges. Both maps are represented exactly by degree-four geometry, up to floating-point error.

Face diagnostics, verification and next decision

Sensor-relevant changes

Strong maps, n=8	Volume error ratio	Face error ratio	Interior jump ratio
Smooth projected fields	1.001	1.002	1
Exactly represented input	5.198e-09	4.058e-09	6.206e-09

Entries are medians of direct/recursive ratios for matched derivative directions: 138 comparisons for sine, shortwave sine and exponential inputs; 46 for the represented quadratic, across P/Q2-P/Q4. Face errors use both one-sided traces; jump RMS uses interior faces, where the analytic jumps vanish. Cases with baseline norms below $1e-9$ are excluded from ratios. These are derivative diagnostics, not full OFDG damping rates.

For the smooth projected inputs, the volume ratios range from 0.983 to 1.373 and the jump ratios from 0.565 to 1.996. Direct differentiation is therefore not uniformly better for those fields. For the exactly represented input, all 46 volume, face and jump comparisons improve. A smaller jump alone does not establish a better shock sensor.

Independent checks and numerical limits

Higher derivatives: 18 first-to-third derivative probes on both maps agree with independently inverted, 65-digit physical finite differences within $1.1e-10$, scaled by $\max(1, \text{reference magnitude})$. Changing the finite-difference step from $1e-5$ to $1e-6$ changes the reference by at most $1e-08$. Finite differences validate the jet method; they are not used by it.

Representation and baseline: polynomial evaluation differs from MFEM by at most $7.3e-14$; physical first derivatives differ by $1.8e-11$ on a $\max(1, \text{magnitude})$ scale. Geometry positions differ by at most $1.4e-13$ after division by h . The 934 matched recursive error norms reproduce the prior accuracy study exactly in the saved CSV values.

Sensitivity: changing input basis changes error norms by at most $1.1e-09$, scaled by $\max(1, \text{analytic derivative L2 norm})$; eight extra evaluation quadrature orders change them by $2e-14$. Direct represented-input relative errors are at most $6.2e-09$ across the suite. Constants have up to $2.1e-08$ absolute derivative L2 error: differentiation amplifies initialization and basis-conversion roundoff. No exact-zero claim is made.

Decision

Retain the existing production method; keep direct differentiation as a promising prototype. It resolves the projection-error mechanism decisively for exactly represented inputs, but does not generally improve derivatives of projected smooth inputs. Before adoption, compare actual OFDG damping rates and PDE behaviour, including shocks, and assess geometry-interface support, cache memory and runtime. The current experiment measures neither stability nor production speed.

Reproduction: docs/direct-derivatives.md; measurements/direct-derivatives/ contains all commands, raw rows, controls and source fingerprints. 50 configurations, 2644 rows, zero failed runs; 52.5 s of monitored numerical runs, one rank/thread on 8 logical processors, peak sampled RSS 21.0 MiB. The independent probes are additional small checks. All monitored cases stayed below 5 minutes and 4 GB.

The previous physical-polynomial reconstruction study is retained separately in measurements/physical-reconstruction/ and docs/physical-reconstruction.md, with its earlier PDF archived. Its pages are omitted here, including from the appendix, to keep this discussion focused.

OFDG on a spherical surface

This experiment evolves a surface-DG solution and applies an OFDG adaptation after each time step. It replaces the FEM-only sphere example in this brief. Both the existing volume implementation and the separate surface-FEM dataset are retained unchanged.

Rigid rotation with an exact solution

On the unit sphere Γ , solve $u_t + \text{div}_\Gamma(v u) = 0$ with $v = (-y, x, 0)$. This velocity is tangential and surface-divergence-free. The exact solution is the initial field evaluated at $(x \cos t + y \sin t, -x \sin t + y \cos t, z)$. Run to $T = \pi/2$, a quarter revolution.

Test	Initial field	Purpose
Smooth rotation	$u_0 = 1 + 0.5x$	Check refinement and the accuracy cost of filtering.
Discontinuous cap	$u_0 = 1$ if $x > 0.5$, otherwise 0	Check oscillation suppression and edge smearing.
Constant control	$u_0 = 1$	Check the surface operator and filter preserve constants.

Surface fluxes and exact curved geometry

Start with octahedral triangles or cubed-sphere quads, then use the pointwise radial map $X = F_h / |F_h|$. All quadrature points lie on the exact sphere. Its 3×2 Jacobian gives tangential gradients through $J(J^T J)^{-1}$. Across each edge, one conservative upwind flux uses v dotted with the tangential outward co-normal. A single flux is added with opposite signs to the neighboring cells.

This radial lift is a custom geometric evaluator in the isolated prototype, not a new general MFEM surface interface. It avoids confusing geometry approximation with the filtering comparison. Solution spaces are P1 triangles and Q1 quads; each family has 96, 384 and 1536 scalar unknowns on the three tested levels.

The first-order surface OFDG definition used here

$$g_{K,a} = \Pi_K(\nabla_\Gamma u_h)_a, \quad A = \max |u_h - \bar{u}|, \quad h_K = (|K|/|\hat{K}|)^{1/2}$$

$$J_{F,0} = |F|^{-1} \int_F |u_h| ds, \quad J_{F,1} = |F|^{-1} \int_F \sum_{a=1}^3 |[g_a]| ds$$

$$\lambda_K = A^{-1} \sum_{F \subset \partial K} \beta_F \left(\frac{J_{F,0}}{2h_K} + \frac{3J_{F,1}}{2} \right), \quad u_h^+ = \bar{u}_K + e^{-\Delta t \lambda_K} (u_h^- - \bar{u}_K)$$

Π_K is the physical surface L2 projection into the original local DG space; $g_{K,a}$ are its three projected tangential-gradient components in fixed Cartesian axes. Bars denote surface means. β_F is the sampled maximum absolute co-normal speed. Face jumps are physical edge averages. A is estimated from volume and edge samples; a relative near-constant guard disables damping below $1e-12$.

The coefficients 1/2 and 3/2 and exponential damping of the nonconstant part are the existing order-one OFDG construction. Surface measures, co-normal speeds and tangential derivatives are the necessary adaptations. The componentwise absolute sum follows the Cartesian convention; no rotation-invariance claim is made. Damping preserves the physical cell mean. No positivity limiter or KXRCF gate is added.

Time integration and support boundary

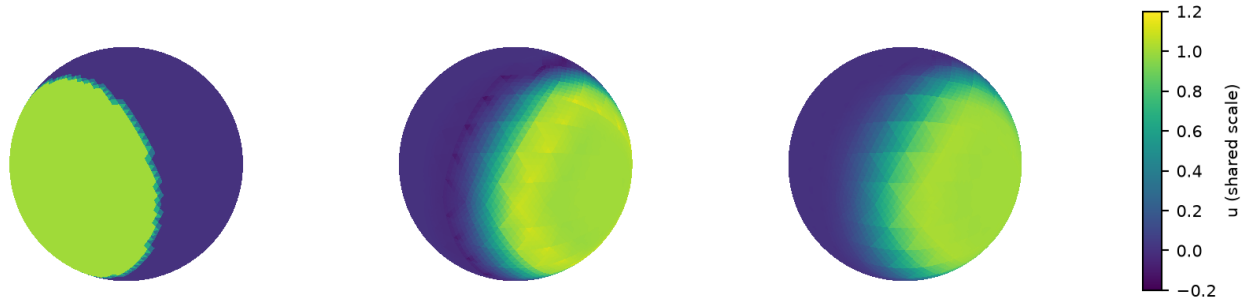
Both methods use the same physical L2 initialization, upwind DG operator and RK4 steps: $\Delta t \leq 0.12 \min(h_K)/3$, shortened to land on T . OFDG is applied once after each complete RK4 step. This is an isolated P1/Q1 prototype, not general surface support in production OFDG. Higher-order surface derivatives, arbitrary surfaces and KXRCF remain unimplemented.

Rotating cap: less oscillation, more smearing

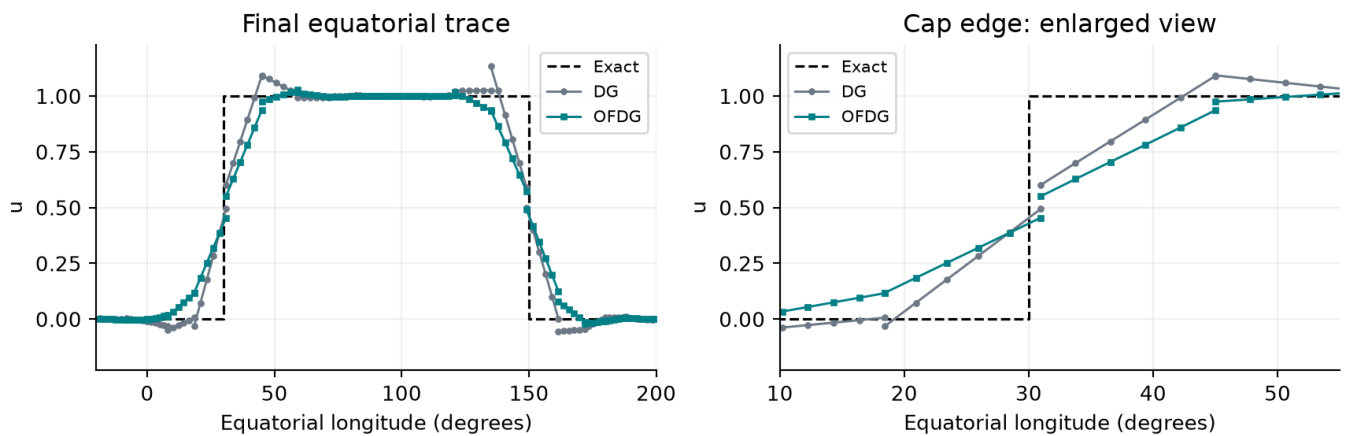
Exact initial cap

DG after quarter rotation

Surface OFDG after quarter rotation



512 triangles, 1536 unknowns. The left panel is the exact initial cap; both methods start from its same unfiltered physical L2 projection, which already has overshoots. The numerical panels show $T=\pi/2$ from the same camera with one shared color scale. Rendering averages values on small display triangles; the tests use quadrature, not image colors.



Equatorial cut at final time: northern one-sided traces, plotted separately within each element to preserve DG jumps. The right panel enlarges the edge at 30 degrees. The exact profile is one between 30 and 150 degrees. The cut does not contain every global extremum.

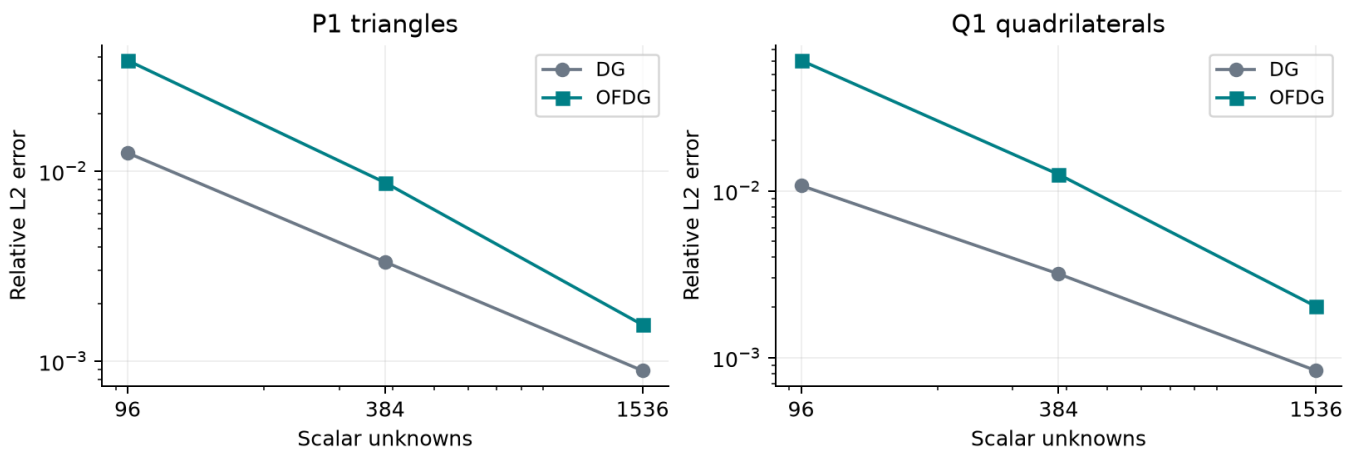
Finest mesh / method	Sampled minimum	Sampled maximum	Relative L2 error
Triangles / DG	-0.1778	1.1728	22.30%
Triangles / OFDG	-0.0440	1.0269	27.40%
Quads / DG	-0.1150	1.1222	21.86%
Quads / OFDG	-0.0222	1.0194	27.80%

What the comparison establishes

The surface OFDG adaptation substantially reduces undershoot and overshoot on both element families, but does not eliminate them. It spreads the discontinuity and increases the L2 error. The result supports a working oscillation-damping mechanism on this surface; it is neither a positivity guarantee nor evidence that filtering improves every error measure.

Errors use the analytic rotated cap and positive physical quadrature of order 60. Raising assembly/input quadrature from 20 to 30 on the finest triangular cap changes reported L2 errors by about 0.053% for DG and 0.042% for OFDG. Integration of a discontinuity is numerical; these values are diagnostic, not certified continuous norms.

Smooth accuracy and validation



Finest mesh / method	Relative L2 error	Observed L2 rate
Triangles / DG	0.08919%	1.90
Triangles / OFDG	0.1554%	2.49
Quads / DG	0.08353%	1.93
Quads / OFDG	0.2026%	2.63

Rates compare the final two refinement levels as the mesh spacing halves. Both methods converge; ungated OFDG has a larger smooth error. Rates above two over this short sequence are not a superconvergence claim. Relative error divides by the full solution norm, including its constant offset.

Conservation, constants and sensitivity

Across the completed suite, total surface mass changes by at most $7.8\text{e-}14$, measured against each run's discrete initial mass. The largest cell-mean change caused by one filter application is $7.8\text{e-}16$. Constant-field final L2 error is at most $6.7\text{e-}15$. The constant-state semidiscrete residual is at most $3.9\text{e-}11$; opposite-side integrated co-normal flux factors agree to $3.3\text{e-}16$.

Halving the finest triangular smooth time step changes DG's L2 error by $2.7\text{e-}09\%$ and OFDG's by 0.16% . Raising smooth assembly/evaluation quadrature by six changes those errors by 0.0111% and 0.0267% . These changes are small relative to the DG/OFDG difference; filter cadence still affects the time-discrete result.

Retained unsuccessful measurement attempt

The initial triangle integration rules included negative weights. They produced an invalid negative squared-error integral in the constant control and unreliable cap error norms. Those measurements are not used here. The source and raw output remain archived in `measurements/sphere-ofdg-initial-quadrature/`. The rerun uses positive tensor/Duffy rules; its checks also reject negative or nonfinite squared-error integrals. No OFDG damping formula was changed to correct that measurement issue.

Decision and remaining work

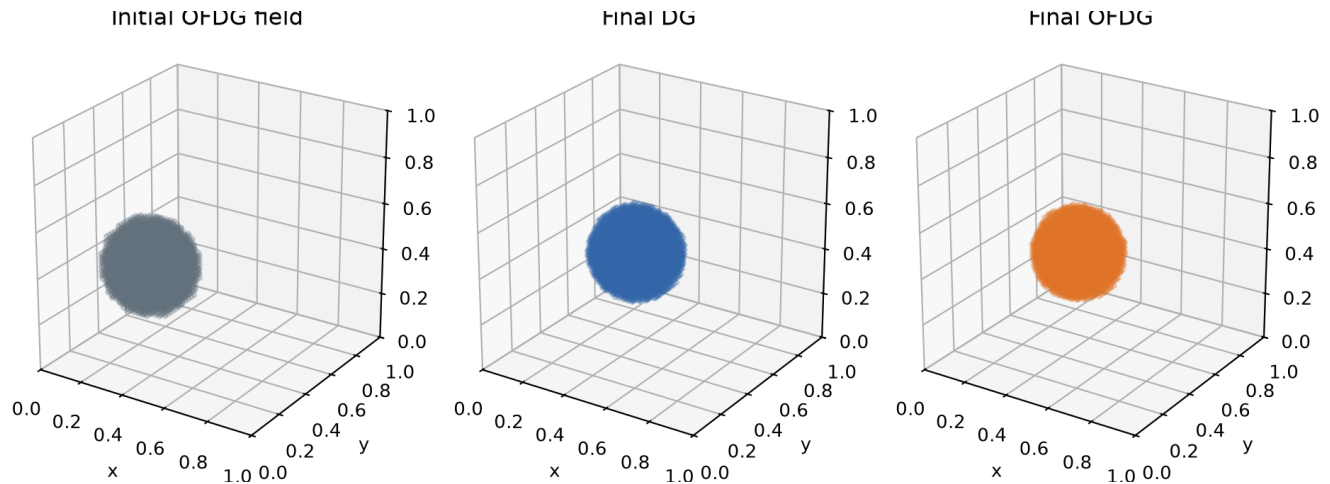
Surface OFDG is now demonstrated in an isolated first-order transport prototype. Keep production volume OFDG unchanged. General integration would require a reviewed surface geometry/flux interface and a deliberate definition of higher-order tangential derivatives. Higher-order surface OFDG and general surface support are out of scope. Retain this first-order example only as an exploratory demonstration.

Resources: 32 final checks, all passed; 49.26 s total monitored numerical time including the first attempt; peak sampled RSS 389.5 MiB. One rank/thread, sequential; five-minute/check and 4 GB ceilings respected. Details: `docs/sphere-ofdg.md`. Commands, source fingerprints, initial/final samples and raw results: `measurements/sphere-ofdg/`. The earlier surface-FEM study is archived separately.

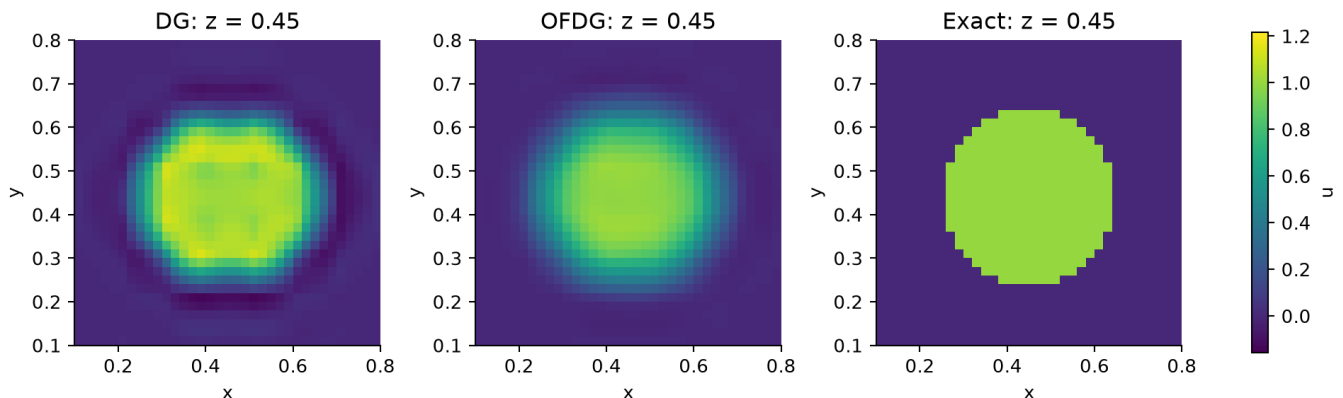
A genuine three-dimensional volume test

The existing production OFDG implementation runs this example without modification. The domain is the volume of the periodic unit cube, meshed with affine hexahedra. This is separate from the first-order spherical-surface prototype; no surface higher derivatives are introduced.

Solve $u_t + 0.5u_x + 0.3u_y + 0.2u_z = 0$. Initially $u=1$ inside a ball of radius 0.2 centered at (0.30,0.35,0.40), and zero outside. At $T=0.3$ its exact center is (0.45,0.44,0.46). The exact field is $u_0(x-vT)$, with periodic wrapping. It varies in all three spatial directions.



Point-cloud views show physical samples where $u > 0.5$, using identical cameras and axes. They locate the transported volume; transparency and this threshold hide small oscillations, which the slices below expose.

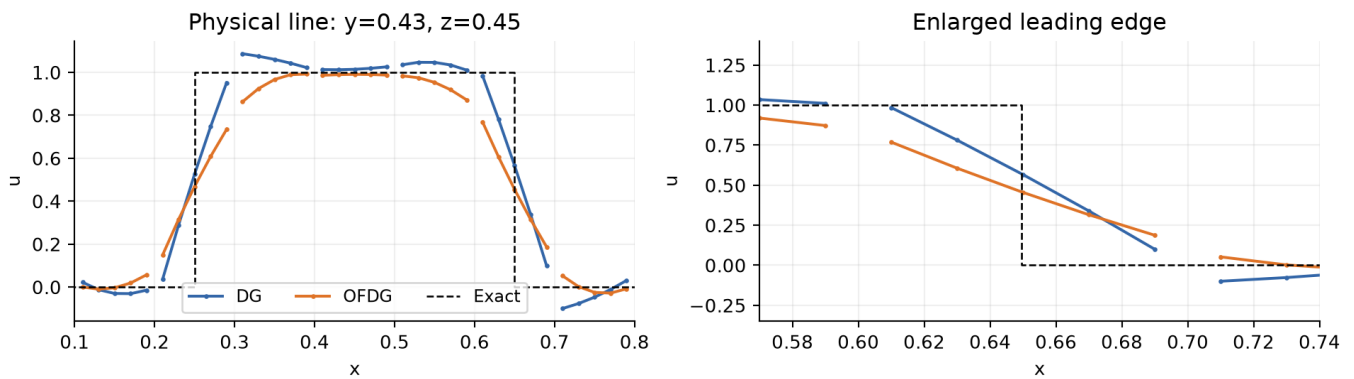


The same physical slice $z=0.45$ for both methods and the exact solution. One color range spans both numerical fields, including excursions outside $[0,1]$. These are polynomial evaluations, not cell averages.

Configuration

Q2 tensor-product DG: 27 scalar unknowns per hexahedron; 6^3 and 10^3 cells (5,832 and 27,000 unknowns). MFEM Rusanov flux, classical RK4, fixed $\Delta t=0.003$, 100 steps. OFDG is applied after every full step without KXRCF or a positivity limiter. Both methods use the same nodal initialization. Initial and final numerical fields and native MFEM mesh/grid-function files are retained.

Oscillation reduction, with visible diffusion



The right panel enlarges the leading discontinuity. Five interior samples per coordinate per cell are exported; line segments stop at cell boundaries. The cut uses exactly $y=0.43, z=0.45$ for both methods and the analytic reference. It is a discontinuously transported interface, not a nonlinear shock.

Pulse / grid	Sampled min	Sampled max	Relative L2
DG / 6^3	-0.3707	1.6498	38.01%
OFDG / 6^3	-0.0399	1.0085	49.73%
DG / 10^3	-0.3026	1.3797	29.82%
OFDG / 10^3	-0.0398	1.0296	39.44%

OFDG reduces both undershoot and overshoot but broadens the interface and increases L2 error. Both pulse errors decrease on the finer grid. The method is not positivity preserving here. Error norms and extrema use order-16 tensor quadrature; integrating a discontinuous reference is approximate, and sampled extrema are not certified global extrema.

Smooth and constant controls

Repeat with $u_0 = 1 + 0.2 \sin(2\pi x) \sin(2\pi y) \sin(2\pi z)$. The exact solution is again translation. Errors below are absolute L2 norms, so the constant background does not dilute their interpretation.

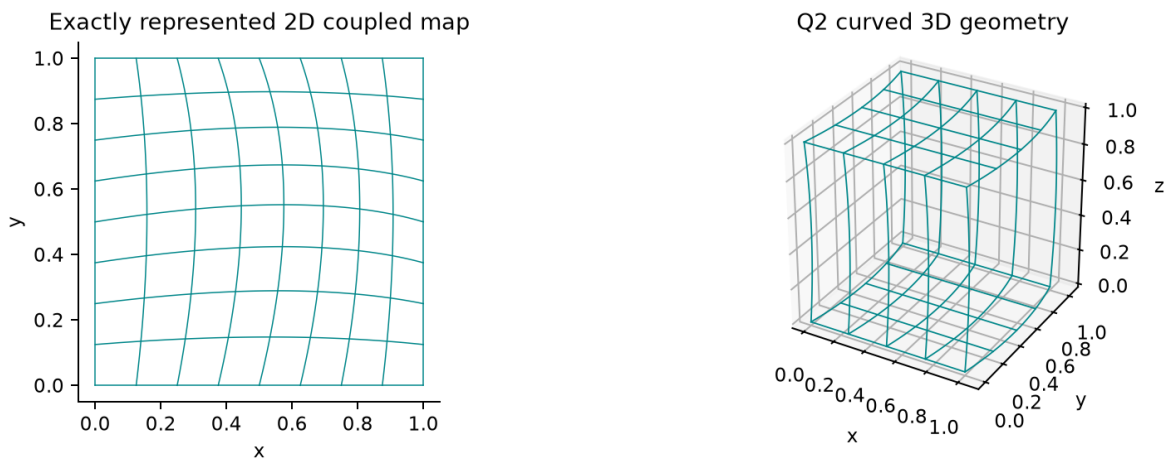
Method	6^3 error	10^3 error	Two-grid rate
DG	0.000864	0.000226	2.62
OFDG	0.00379	0.000611	3.57

Constant OFDG control: final L2 error $2.6e-16$. Largest absolute periodic mass drift: $1.9e-14$, relative to each discrete initial mass. Halving the fine pulse time step changes OFDG L2 error by 0.0534%. These are small checks, not an asymptotic convergence or performance study.

Scope and resources: affine hexahedral scalar transport only; this showcase does not validate curved 3D, tetrahedra or 3D Euler. 10 runs, 52.1 s monitored compute; peak sampled process-tree memory 80.9 MiB. One MPI rank, one library thread; sequential runs. All checks passed. Reproduction: [docs/volume-showcase.md](#); raw data and source fingerprints: [measurements/volume-showcase/](#).

The mesh itself supplies the mapping derivatives

Question: can finite element coordinate fields replace the analytic mapping formulas needed by the earlier direct-derivative experiment? **Result:** yes for the tested polynomial volume meshes through third order. The new method agrees with analytic-map differentiation within the prescribed scaled tolerance.



Left: the coupled square map has genuinely curved element edges. Right: $x=s+0.2t(1-t)$, $y=t+0.15r(1-r)$, $z=r$, with determinant one and curved faces. Geometry is represented exactly in the selected FE spaces in exact arithmetic; the plots show the defining polynomial meshes.

A shared evaluator, with independent geometry and solution spaces

Read the geometry basis and coordinate coefficients from MFEM (GetFE and GetPointMat). Treat x , y and z as components of an FE field. Evaluate their reference-coordinate Taylor expansions with the same polynomial machinery as the solution; invert the local geometry series and compose the original solution with it. No derivative is projected along this new path. Existing geometry coefficients are reused, so no mesh projection is introduced here.

Method	Source of geometry / derivative construction
Recursive	Unchanged production projected physical derivative matrices; repeat in canonical x , y , z order.
Analytic direct	Original mapped solution differentiated using explicit fixture mapping formulas.
FE geometry	Same mapped solution differentiated using only stored geometry coefficients and basis.

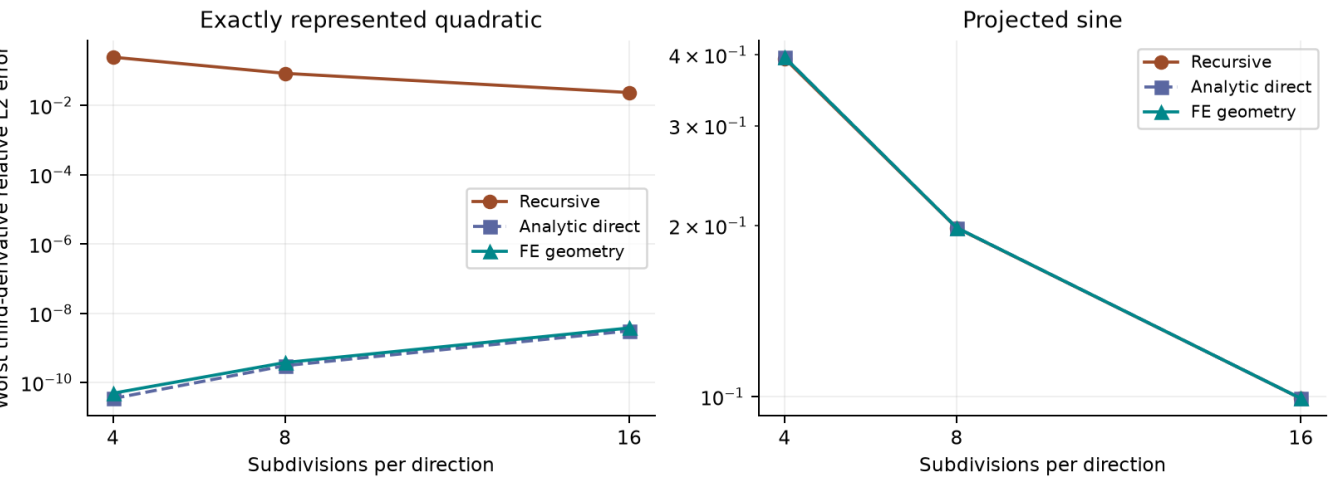
The 2D suite repeats all 50 earlier configurations. Nine small 3D cases compare Q2/Q3 solutions on 2^3 and 4^3 cells with Q2 geometry, including affine and quadrature controls. All methods receive identical input coefficients and evaluation points. Geometry degree and solution degree are independent.

Scope: an isolated benchmark, not an OFDG or PDE modification. Production sources, existing benchmarks and thesis content are preserved. Agreement validates the representation-based route, not improved PDE accuracy or arbitrary geometry support.

Three methods on the same represented mesh and solution

Entries are the **worst relative volume L2 error across third derivatives**. The maximizing derivative may differ between methods. 2D: $n=8$, P/Q3, strong curvature. The exactly represented quadratic removes input approximation error in exact arithmetic; the sine is projected into the same solution space.

Input / space	Recursive	Analytic direct	FE geometry
Exact quadratic / Q3	0.0846	3.05e-10	3.76e-10
Projected sine / Q3	0.198	0.198	0.198
Exact quadratic / P3	0.0722	7.89e-10	4.62e-09
Projected sine / P3	0.428	0.435	0.435



Values are fractions, not percentages. Plots show Q3 only. For exactly represented inputs the direct methods remove intermediate projection error, but retain amplified floating-point and basis-conversion error. For the sine, differentiated input approximation dominates: removing operator projection does not guarantee lower error against the analytic field. Overlapping direct-method curves are expected.

Independent checks and preserved baselines

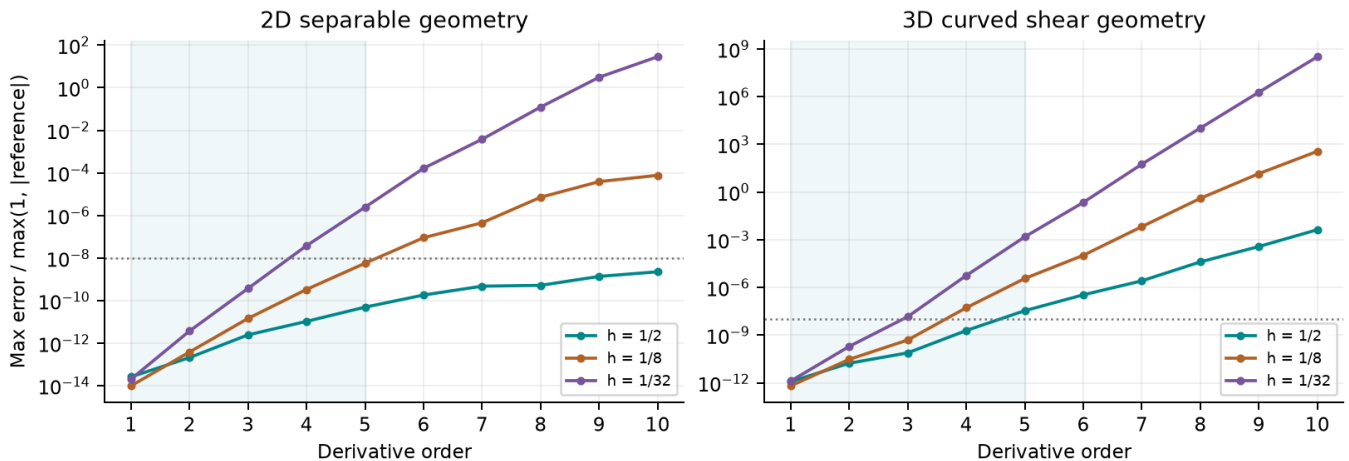
Check	Largest observed discrepancy
Earlier baseline rows	2644 matched; 0 change in tested norms
FE vs analytic direct derivatives	1.3e-09 on element-scaled derivative magnitudes
Geometry derivatives / shared faces	2.6e-11 / 3e-13
Independent physical finite differences	46 probes; scaled error 1.1e-10
Finite-difference step sensitivity	1e-08 (steps 1e-5 and 1e-6)

Positions, Jacobians, inverse residuals, constants, affine polynomials, face traces, basis changes and quadrature checks passed. Constant derivative L2 errors reach 2.1e-08; there is no exact-zero claim. One-sided face errors and interior jumps are retained in the data, but are not complete OFDG damping rates.

Curved 3D works; high orders expose numerical limits

3D Q3, n=4: third derivatives	Recursive	Analytic direct	FE geometry
Exactly represented polynomial	1.02e-06	7.21e-10	7.21e-10
Projected sine	0.471	0.472	0.472

Worst relative volume L2 error among nonzero third derivatives. All 19 derivative multi-indices through order three are evaluated; zero-reference derivatives use absolute errors. The exact pullback is $s^2+t^2+r^2+str$. The sine is $\sin(2x+3y+z)$.



Higher-order stress test: Q5 nodal inputs on representative elements of size $h=1/2$, $1/8$ and $1/32$; three points, all pure and mixed derivatives through order ten, affine and curved controls. References use 90-digit explicit-inverse formulas/polynomials, checked at 120 digits. Curves show $\max |error|/\max(1, |reference|)$ across points and directions; this is not uniformly a relative error.

Curved probe	h	Order 5	Order 10
2D	1/2	4.94e-11	2.27e-09
2D	1/32	2.45e-06	28.6
3D	1/2	3.43e-08	0.00426
3D	1/32	0.00151	3.24e+08

Interpretation: the construction extends to any finite order; accurate evaluation in ordinary precision does not. Differentiation amplifies nodal initialization, polynomial conversion, geometry and arithmetic errors, particularly on small elements. Fifth-order accuracy is useful in several probes but is not uniformly near roundoff; tenth order can fail badly. The Q5 input is exactly representable mathematically, not immune to floating-point errors. Analytic-map direct evaluation shows essentially the same 3D $h=1/32$ errors (0.00151 at order five; 3.24e8 at order ten).

Order-five and order-ten specializations agree through order five within $7.9e-16$ on a $\max(1, \text{magnitude})$ scale. The smaller specialization limits work for ordinary use. These probes identify accuracy limits; they do not establish production performance or high-order stability.

Reproduce: docs/fe-mapping-derivatives.md. 59 main cases plus 14 high-order cases; 5,505 main and 39,150 high-order rows. Monitored compute 319.4 s; peak sampled memory 27.3 MiB. One rank/thread, sequential measured cases; independent reference analysis and compilation are additional.

Reuse the affine machinery; differentiate by the chain rule

Two additional variants reuse the existing basis-independent reference derivative matrices for both the solution and the coordinate fields. They differentiate the inverse Jacobian algebraically, then apply the chain rule repeatedly. No analytic map or inverse-map Taylor expansion enters the new evaluator.

Stage	Construction
Reference derivatives	Call the unchanged production assembler $D_a = V^{-1}G_a$; repeatedly apply D_a to coefficient vectors. Geometry and solution may have different bases and degrees.
Inverse Jacobian	Scale $K = J/h$ and $B = K^{-1}$. Differentiate $KB = I$ and solve for each higher derivative of B . Preserve matrix multiplication order.
Physical derivatives	Apply $h \partial/\partial x_i = \sum_a B_{ai} \partial/\partial \xi_a$, retaining product-rule terms through the requested order. Divide by h^m at order m .
Optional centering	Subtract $c \cdot q$ before differentiation, with $Vq = 1$. Apply separately to each coordinate and solution component; this remains valid when q is not an all-ones vector.

The graph driver is adapted locally to arbitrary requested order, including structural P/Q polynomial zeros. The production matrix assembler is reused literally. This experiment makes no production-interface or damping changes.

Validation	Largest observed discrepancy
MFEM Jacobian / physical first derivative	3.6e-12 / 2e-11
Differentiated inverse identity $KB = I$	4.9e-15
Fifth-order polynomial basis reproduction	1.6e-10 across 24 basis/centering controls
Preserved older rows	5,505 main + 39,150 high-order values; zero measured change

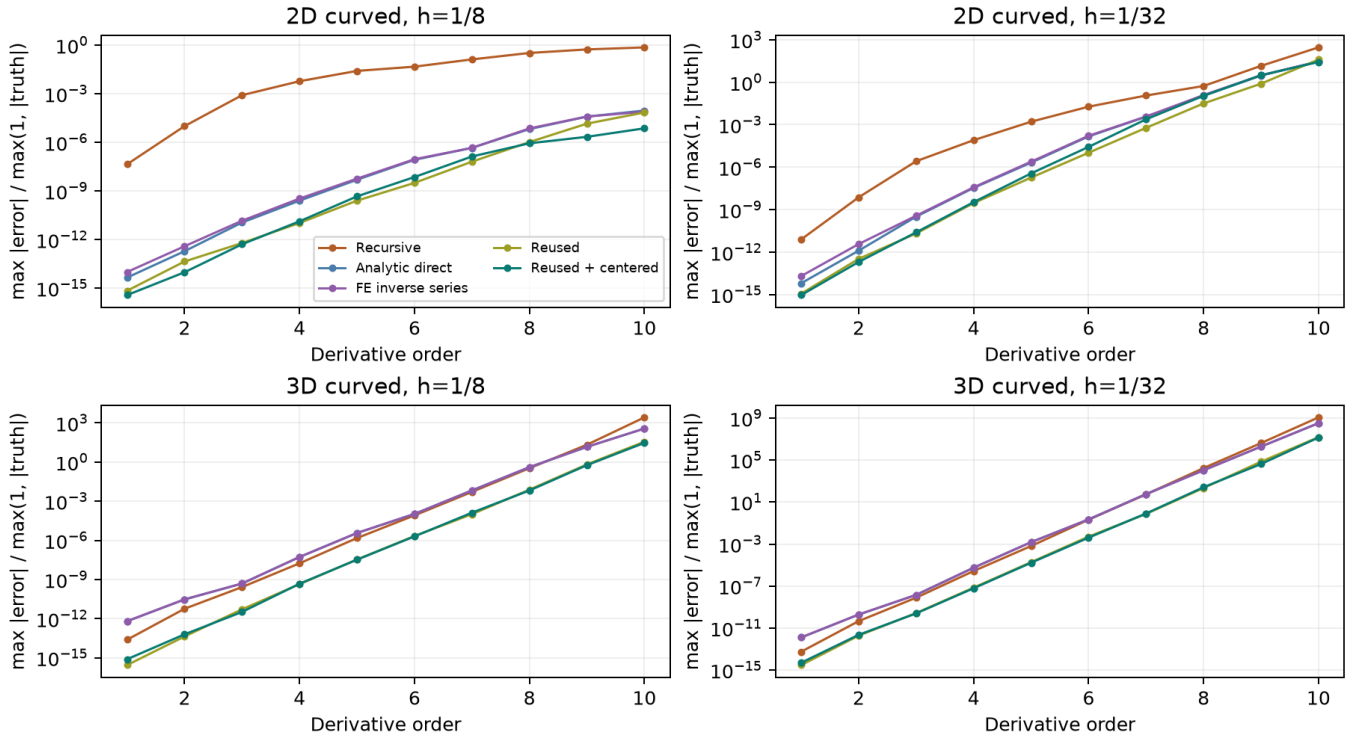
Basis controls include Gauss-Lobatto, Gauss-Legendre and Bernstein, plus scaled bases whose constant coefficients are not all ones. The largest constant-vector representation error is 1.4e-15.

Third-order relative L2	Recursive	Analytic	FE series	Reuse	Centered
2D Q3 exact input	0.085	1.1e-09	1.1e-09	1.1e-09	1.1e-09
2D P3 exact input	0.072	1.7e-09	7.2e-09	8.8e-09	2.3e-09
3D Q3 exact input	1e-06	7.2e-10	7.2e-10	7.2e-10	7.2e-10
3D Q3 projected sine	0.47	0.47	0.47	0.47	0.47

Worst nonzero-reference third derivative across each selected group (2D includes the tested curvature amplitudes). Values are fractions. The sine is limited by its FE input approximation; direct evaluation cannot recover derivatives discarded by that approximation.

Fifth-order accuracy improves; tenth order remains limited

All five methods receive the same Q5 coefficients and mesh. Curved local-element probes use three points and every derivative multi-index through order ten. The ideal-field reference is computed with 90-digit explicit-inverse formulas.



Curved, $h=1/32$	Recursive	Analytic	FE series	Reuse	Centered
2D / order 5	0.0016	2.1e-06	2.5e-06	1.8e-07	3.6e-07
3D / order 5	0.00065	0.0015	0.0015	1.9e-05	1.7e-05
2D / order 10	2.9e+02	29	29	40	27
3D / order 10	1.2e+09	3.2e+08	3.2e+08	1.5e+07	1.4e+07

At $h=1/32$, order-five **sampled relative L2** falls from $3.34 \cdot 10^{-8}$ (FE series) to $3.84 \cdot 10^{-9}$ (reuse) in 2D, and from $5.65 \cdot 10^{-6}$ to $1.18 \cdot 10^{-7}$ in 3D. Reuse gives maximum h^5 -scaled absolute errors of $3.55 \cdot 10^{-14}$ and $5.63 \cdot 10^{-13}$, respectively.

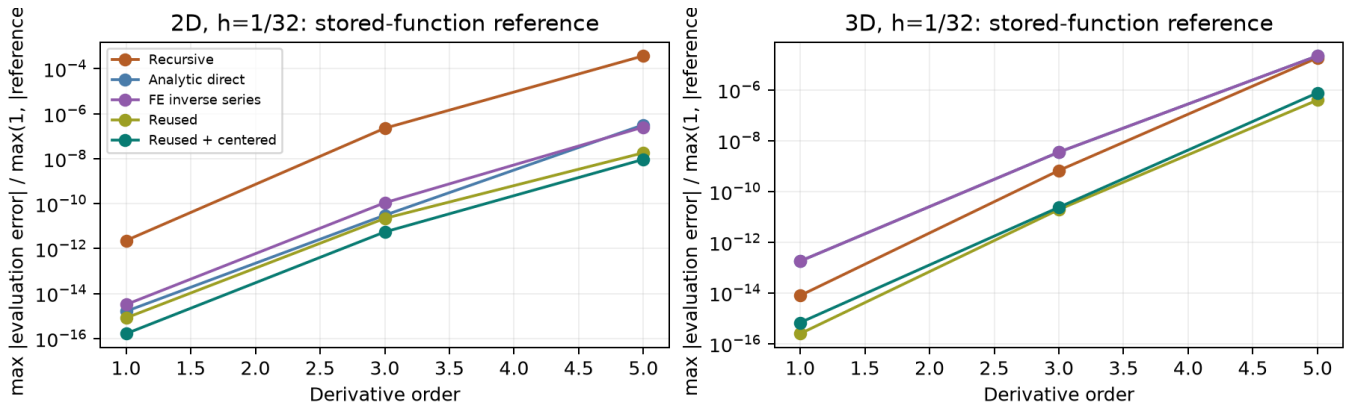
Plot/table metric: $\max |error| / \max(1, |ideal\ derivative|)$, over points and directions. This is a mixed absolute/relative scale, not uniformly a relative error. The data also contain absolute errors, relative errors, order-group discrete relative L2, and h^m -scaled errors. No integration weights enter these local stress norms.

Reusing the native derivative matrices avoids the earlier monomial-conversion path. Centering reduces cancellation in some cases, especially constants, but does not improve every derivative. Both approaches retain coefficient roundoff, differentiation conditioning and the physical h^{-m} amplification. Fifth-order results must be assessed by dimension and mesh size; tenth order has no general accuracy guarantee.

Requesting capacity five instead of ten changes the shared outputs by at most $7.9e-16$ on a $\max(1, \text{magnitude})$ scale. The order-five path therefore avoids unnecessary high-order work. These are diagnostic measurements, not an optimized production implementation.

Separate evaluator error from the represented input

A second, independent reference differentiates the **same stored FE function and geometry**: reconstruct their tensor Lagrange polynomials from exact binary64 coefficients/nodes, invert the map with 90-digit Newton iteration, then take high-precision physical finite differences. No benchmark chain-rule or inverse-series arithmetic is shared.



Order 5, h=1/32	Analytic	FE series	Reuse	Centered
2D evaluation error	1.5e-06	1.2e-06	4.2e-07	8.9e-08
2D representation error	1.9e-07	1.9e-07	1.9e-07	1.9e-07
3D evaluation error	8.5e-05	8.5e-05	2.7e-06	7.6e-07
3D representation error	1.4e-06	1.4e-06	1.4e-06	1.4e-06

Table: maximum absolute errors over the selected fifth-order directions. Representation error is shared because the input is shared. Each column can attain its maximum in a different direction; these maxima must not be added. The plot uses $\max(|\text{evaluation error}|, \max(1, |\text{stored reference}|))$.

Measure	What it diagnoses
Total error	Computed derivative minus the ideal analytic-field derivative. This is the final accuracy experienced by the user.
Evaluation error	Computed derivative minus the high-precision derivative of stored u_h on stored F_h . Tests the differentiation machinery.
Representation error	Stored-function derivative minus the ideal-field derivative. Here the spaces can represent the fixtures exactly, so coefficient/geometry roundoff remains.
Relative and element-scaled errors	Relative L2 normalizes by reference norm; use absolute error for zero truth. h^m -scaled error exposes amplification but does not replace physical error.

Accuracy limit: selected fifth-order evaluation errors exceed 10^{-7} on the normalized scale (up to $7.6 \cdot 10^{-7}$ for centered reuse). The stable reference steps identify a numerical limitation, not stencil uncertainty.

20 independent pure/mixed probes, orders 1, 3 and 5, in 2D/3D at $h=1/8$ and $1/32$. Steps $h \cdot 10^{-4}$ and $h \cdot 5 \cdot 10^{-5}$ differ by at most $1.8e-17$ on a $\max(1, \text{reference})$ scale. This validates selected directions, not every tenth-order output.

Reproduce: docs/reused-derivatives.md. 73 monitored configurations; 9,175 main and 65,250 high-order rows. Sequential one-rank/one-thread compute: 409.2 s; peak sampled memory 34.0 MiB. Initialization and evaluation timings are retained for the combined diagnostic program. Reference analysis and compilation are additional.